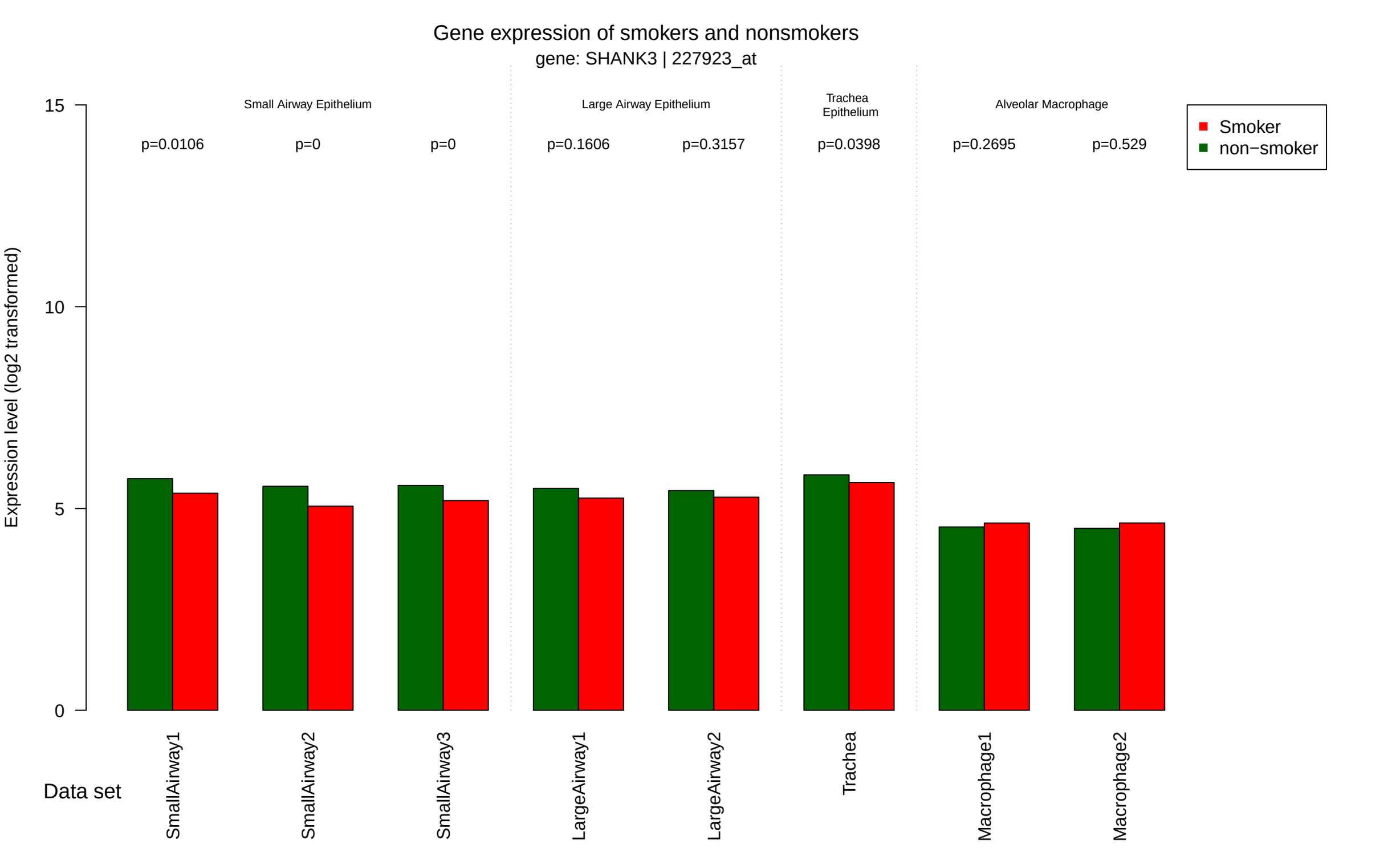
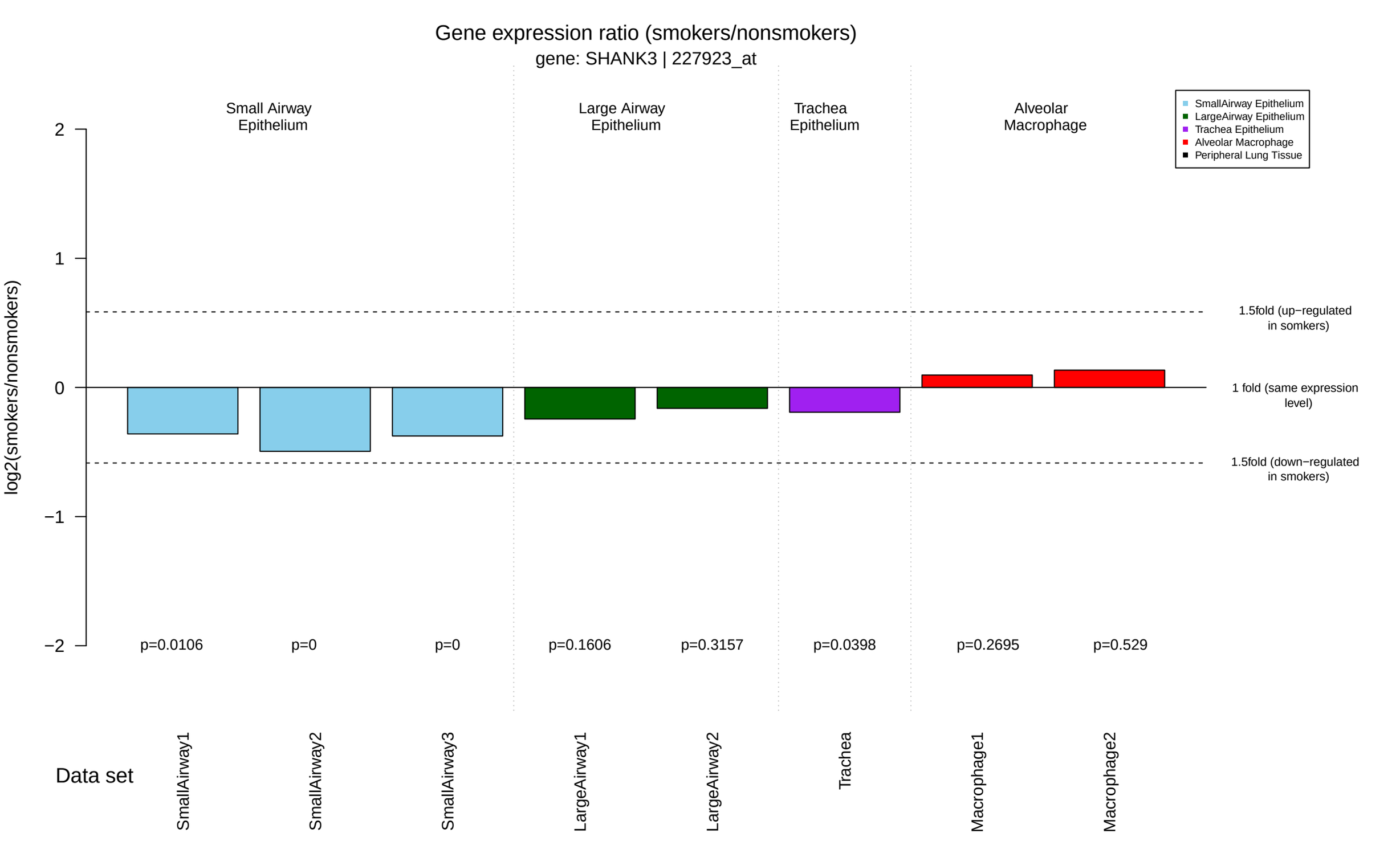
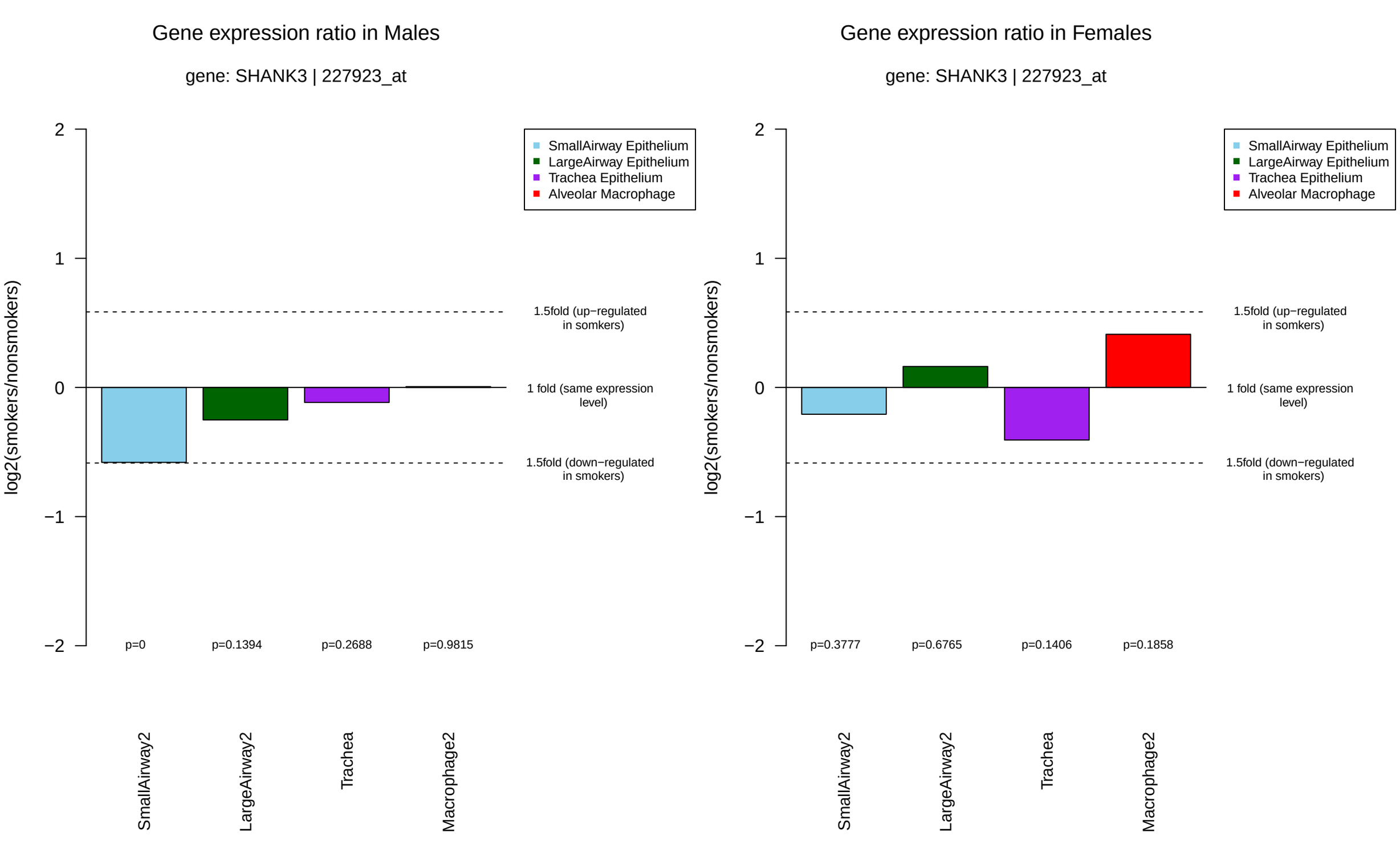
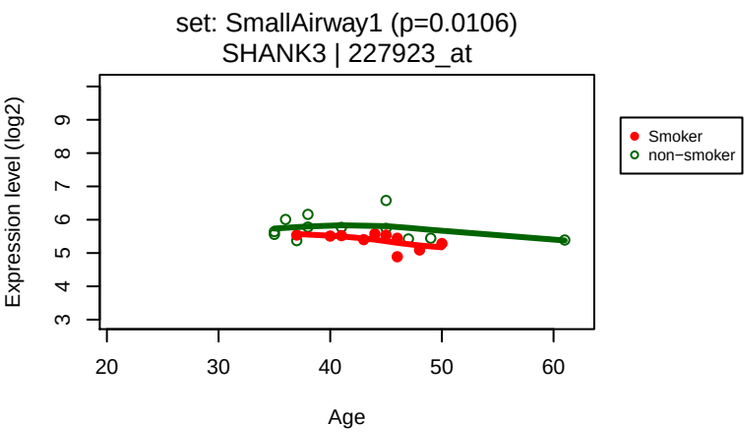
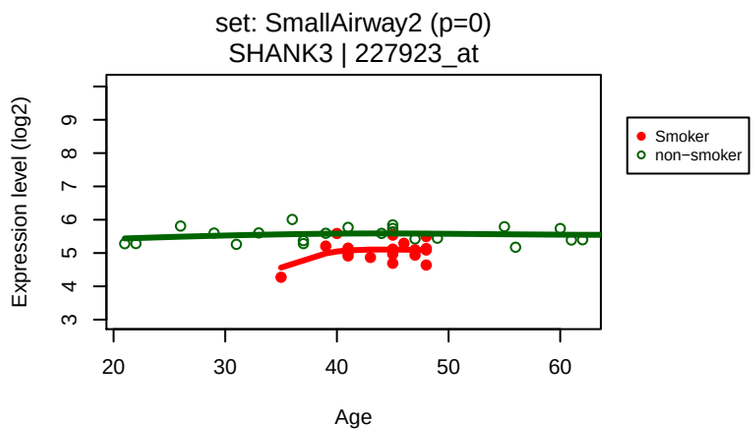
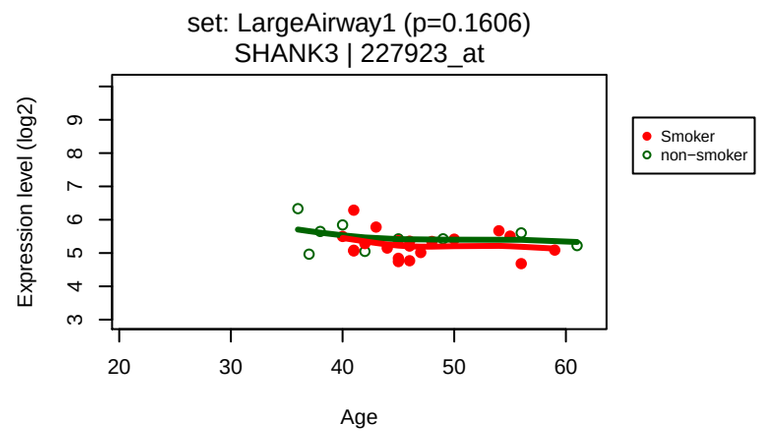

Gene expression of smokers and nonsmokers
gene: SHANK3 | 227923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0106
p=0
p=0
p=0.1606
p=0.3157
p=0.0398
p=0.2695
p=0.529
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

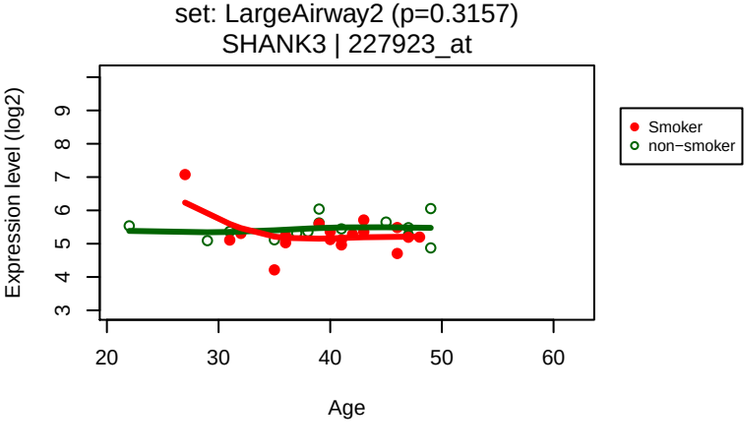

Gene expression ratio (smokers/nonsmokers)
gene: SHANK3 | 227923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0106
p=0
p=0
p=0.1606
p=0.3157
p=0.0398
p=0.2695
p=0.529
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

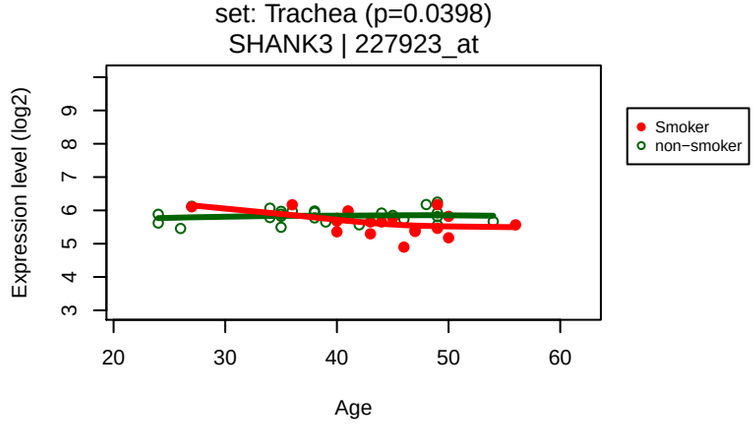

Gene expression ratio in Males
gene: SHANK3 | 227923_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0
p=0.1394
p=0.2688
p=0.9815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SHANK3 | 227923_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3777
p=0.6765
p=0.1406
p=0.1858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

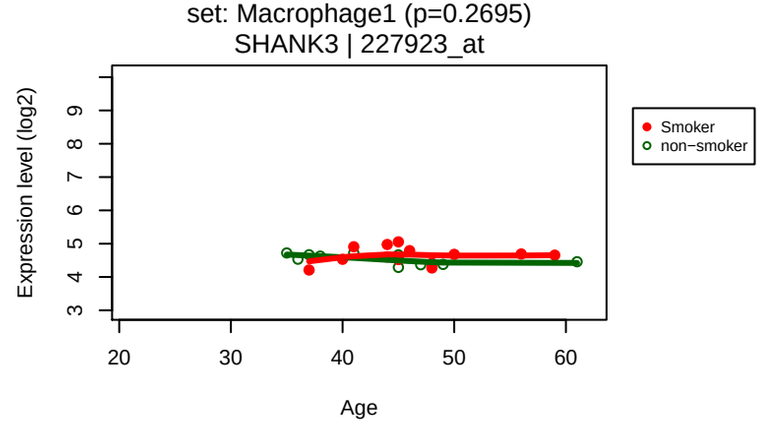

set: SmallAirway1 (p=0.0106)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker

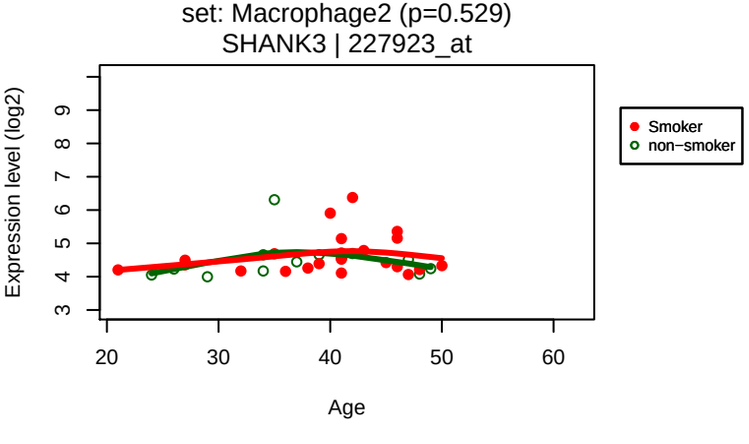

set: SmallAirway2 (p=0)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker

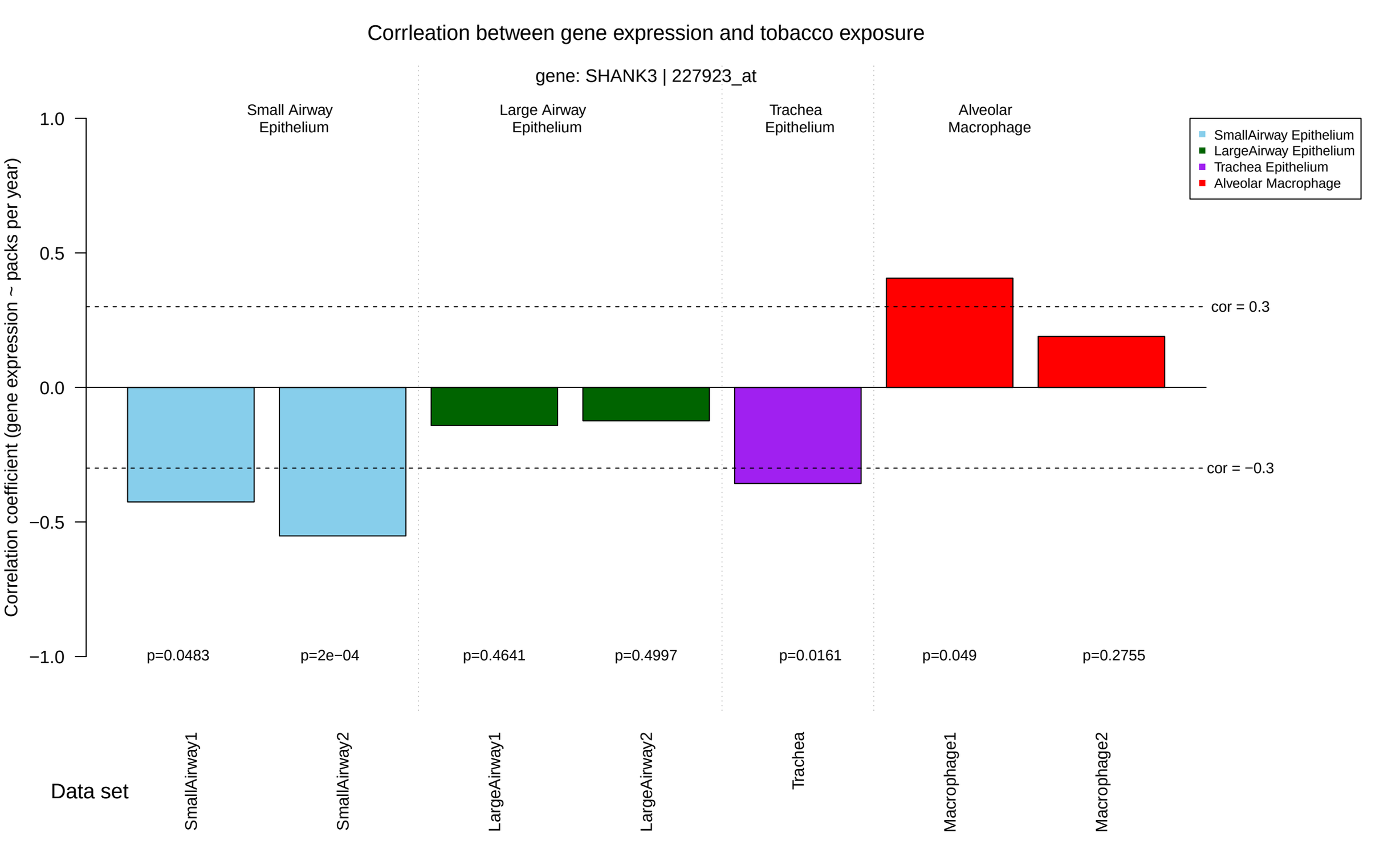

set: LargeAirway1 (p=0.1606)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3157)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0398)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2695)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.529)
SHANK3 | 227923_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SHANK3 | 227923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0483
p=2e−04
p=0.4641
p=0.4997
p=0.0161
p=0.049
p=0.2755
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2